

Project Design Phase-II

Data Flow Diagram & User Stories

Date	03 October 2026
Team ID	SWTID-2026-5030
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

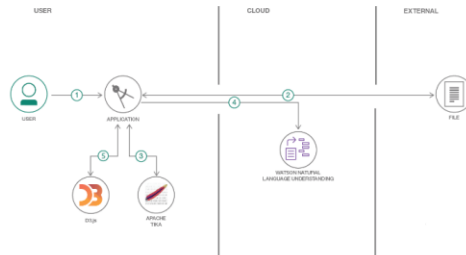
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

1. **User Request Initiation:** Employee accesses the Service Catalog via ServiceNow / Service Portal; selects standard laptop configuration with auto-populated user and department metadata.
2. **Automated Approval Routing:** Flow Designer triggers conditional approval workflows to Reporting Manager and IT Procurement Lead via Email, Teams/Slack, or ServiceNow Portal.
3. **Catalog Task Creation & Assignment:** Upon approval, Flow Designer creates a Catalog Task ("Laptop need to Configured"), setting Assignment Group to "Hardware" and status to "Approved".
4. **Vendor PO Generation & Fulfillment:** Automated REST API call sends the order payload directly to the external hardware vendor endpoint, who then ships hardware to Desktop Support.
5. **Delivery, Imaging & Setup:** Desktop Support Engineer receives laptop, flashes base OS image, and dispatches device to employee.
6. **Asset Lifecycle & CMDB Integration:** ServiceNow CMDB automatically updates hardware state from "In Stock" to "In Use" and triggers an automated user satisfaction survey.

Example: [\(Simplified\)](#)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Employee - Web/Mobile User)	Request Initiation	USN-1	As an employee, I can select and order a standard laptop configuration from the Service Catalog, so that I don't need manual email exchanges.	Standard laptop catalog item is accessible; form auto-populates user & department metadata; request submits successfully.	High	Sprint-1
Customer (Employee - Web/Mobile User)	Request Tracking	USN-2	As an employee, I will receive confirmation and status notifications regarding my laptop request, so that I can track delivery timeline.	Automated email notifications sent upon request submission, manager approval, and device dispatch.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer Care / Reporting Manager	Approval Management	USN-3	As a reporting manager, I can review and approve/reject laptop procurement requests directly from my notifications, so that approval bottlenecks are avoided.	Approval tasks contain full employee and specs metadata; approval status updates in real-time.	High	Sprint-1
IT Procurement Lead	Flow Designer Automation	USN-4	As an IT procurement lead, I want Flow Designer to automatically generate catalog tasks upon approval, so that tasks are instantly assigned without human intervention.	Flow "Standard Laptop Task" executes automatically post-approval; sets short description "Laptop need to Configured" and group "Hardware".	High	Sprint-1
Hardware Team / Desktop Engineer	Fulfillment & Staging	USN-5	As a hardware support engineer, I can view assigned catalog tasks in the Hardware queue, so that I can flash the OS image and stage laptops promptly.	Catalog task appears in Hardware group queue with status "Approved"; task can be marked complete upon staging.	High	Sprint-2
Administrator / Asset Manager	Asset Lifecycle & CMDB Sync	USN-6	As an IT asset manager, I want the CMDB database to automatically update asset status to "In Use" upon deployment, so that inventory data remains accurate.	Asset record state transitions from "In Stock" to "In Use"; automatic satisfaction survey triggers upon delivery.	Medium	Sprint-2